

Unit 5 Part II

3D Printing

Prepared by

Dr. Pragya Sharma
Asst. Professor
Department of Mechanical Engineering
ABESEC

1. Additive Manufacturing

ISO/ASTM 52900 define additive manufacturing (AM) as the process of joining materials to make parts from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing and formative manufacturing methodologies.

As per the above definition the following are the mandatory condition for a process of qualify as AM

1. It should be a process of joining of material.
2. It should be starting from three-dimensional (3D) model data.
3. It should have preferably layer by layer build up approach for manufacturing.
4. It should not follow subtractive manufacturing methodologies.
5. It should not follow formative manufacturing methodologies.

- **Additive Manufacturing (AM) = umbrella term**
Covers all processes where objects are made layer-by-layer by adding material.
- **3D Printing = popular subset / informal term**
Originally referred to simpler, desktop-scale or prototyping processes, it is now commonly used to refer to all AM technologies, even in industrial settings.

2. Introduction to 3D Printing

Definition:

3D Printing, also known as Additive Manufacturing (AM), is a process of creating three-dimensional objects layer by layer from a digital 3D model. Unlike traditional manufacturing (subtractive), which removes material, 3D printing adds material to build the part.

The creation of a 3D printed object is achieved using additive processes. In an additive process, an object is created by laying down successive layers of material until the entire object is created. Each of these layers can be seen as a thinly sliced horizontal cross-section of the eventual object.

The process starts with making a virtual design of the object that has to be created. This virtual design is made in a CAD (Computer Aided Design) file using a 3D modelling program (new object) or using a 3D scanner (existing object). Slicing software slices the final model into hundreds or thousands of horizontal layers. When this prepared file is uploaded in the 3D printer, the printer creates the object layer by layer. The 3D printer reads every slice (or 2D image) and proceeds to create the object, blending each layer with no sign of the layering visible, resulting in one three-dimensional object.

3. History of 3D Printing

- **1980** – Dr. Kodama files first patent for Rapid Prototyping (RP).
- **1983–1986** – Chuck Hull invents Stereolithography (SLA) and co-founds 3D Systems.

- **1987–1989** – Launch of SLA-1; Carl Deckard patents Selective Laser Sintering (SLS); Scott Crump patents Fused Deposition Modeling (FDM).
- **1990s** – Expansion into industrial prototyping and tooling; companies like SolidScape, ZCorp, and Arcam emerge.
- **2000–2005** – Growth of high-end applications in aerospace, medical, and automotive; SLM, EnvisionTec, ExOne introduced.
- **2004–2007** – RepRap project begins, promoting open-source, self-replicating 3D printers.
- **2009** – First commercial RepRap-based printers launched; MakerBot enters market.
- **2012** – Affordable desktop resin printers like B9Creator and Form 1 launched via Kickstarter.
- **2013** – MakerBot acquired by Stratasys; significant media coverage and industry growth.
- **Present** – Widely adopted in aerospace, medical, automotive, and consumer products, driving Industry 4.0 advancements.

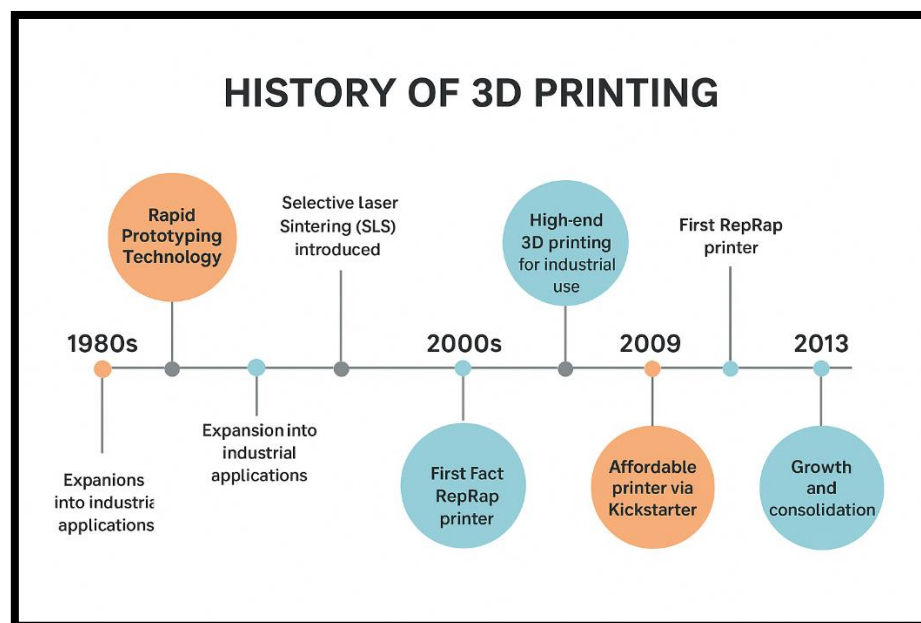
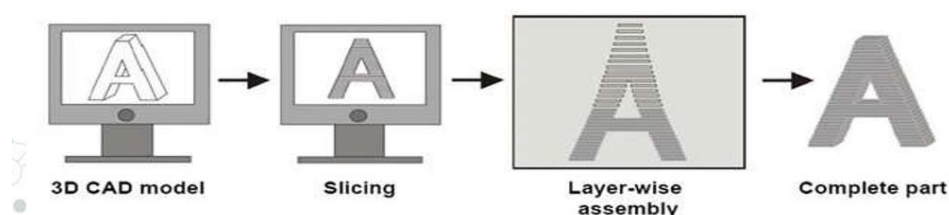


Figure- History of 3D Printing

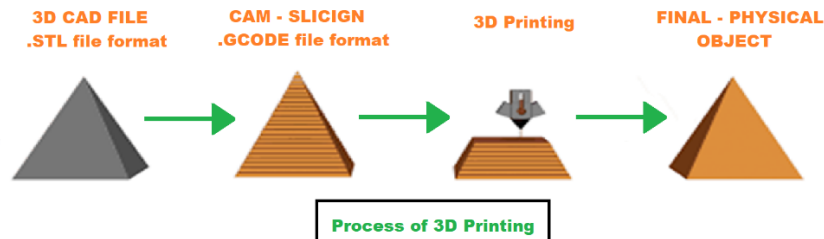
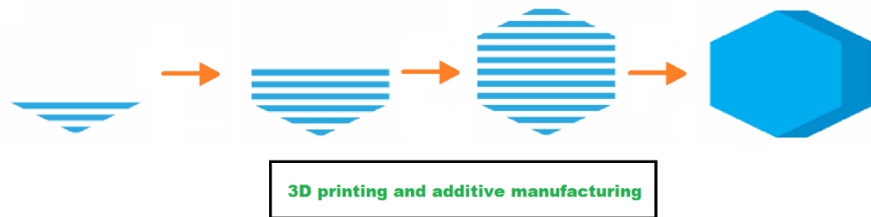
There are 2 types of software required to create a 3D printed model-

1. 3D design software where one creates a 3D design file of the object one wishes to 3D print. e.g TinkerCAD, Google Sketchup, AutoCAD, Fusion 360 etc.
2. 3D slicing software where the output file of the 3D object from 3D design software is built and sliced for 3D printing. e.g Cura, GrabCAD, MakerBot, etc.



What is 3D Printing?

➡ Structuring a three-dimensional object in its physical configuration from its digital form.



4. 3D Printing Process Workflow

1. Design (CAD Modeling)

Create a 3D model using software like Fusion 360, SolidWorks, or AutoCAD.

2. Conversion to STL

Export the model into .STL format (triangulated mesh file).

3. Slicing

Use slicing software (e.g., Cura, PrusaSlicer) to divide the model into layers and set printing parameters.

4. Printing

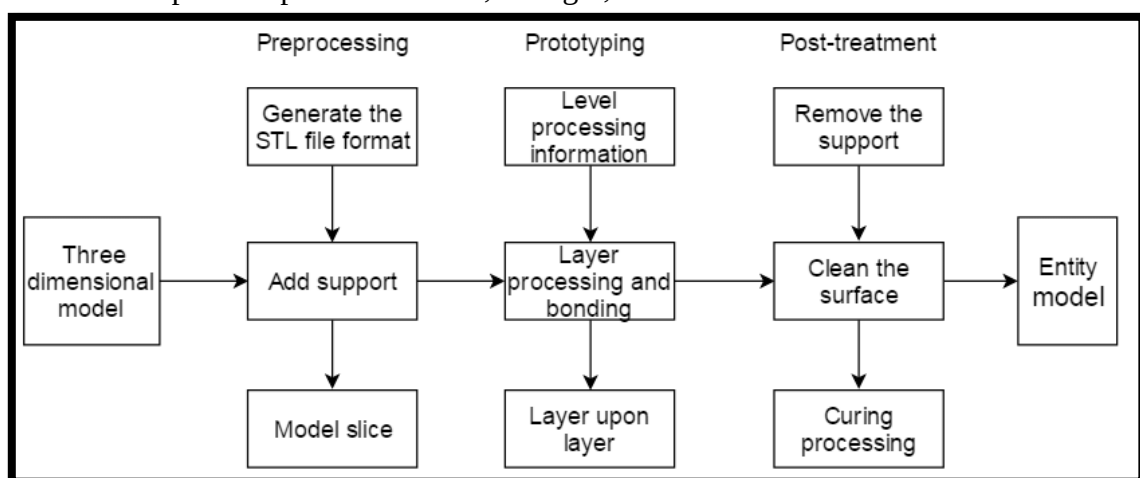
Send sliced file to the printer and start the printing process.

5. Post-processing

Includes support removal, sanding, curing (for resin prints), or heat treatment (for metals).

6. Testing and Quality Check

Inspect the part dimensions, strength, and surface finish.



5. Types of 3D Printing Technologies

Technology	Process Description	Common Materials	Advantages	Limitations	Applications
FDM (Fused Deposition Modeling)	Thermoplastic filament is heated, extruded, and deposited layer by layer through a nozzle.	PLA, ABS, PETG, Nylon, TPU	Low cost, easy to use, wide material availability	Visible layer lines, lower resolution, weaker parts in Z-direction	Prototypes, functional parts, hobby printing
SLA (Stereolithography)	UV laser cures liquid photopolymer resin layer by layer in a resin vat.	Photopolymer resins	High precision, smooth surface finish, fine details	Brittle parts, limited mechanical strength, post-curing required	Dental models, jewelry, engineering prototypes
SLS (Selective Laser Sintering)	Laser sinters powdered thermoplastic material to create solid layers without support structures.	Nylon (PA12), TPU, Polyamide, Glass-filled nylon	Strong, durable, complex geometries, no supports needed	Rough surface finish, expensive machines	Functional prototypes, aerospace, automotive parts
DLP (Digital Light Processing)	Uses a digital light projector to cure resin layer by layer, similar to SLA but faster.	Photopolymer resins	Faster than SLA, high resolution, smooth finish	Resin cost, brittle parts, limited build volume	Dental aligners, jewellery, high-detail models
DMLS / SLM (Direct Metal Laser Sintering / Selective Laser Melting)	High-power laser melts metal powder layer by layer to form fully dense metal parts.	Stainless steel, titanium, aluminum, cobalt-chrome, tool steels	High strength, fully functional metal parts, complex designs	Expensive, requires post-processing and support removal	Aerospace, automotive, medical implants, tooling

Binder Jetting	Liquid binder selectively deposits to bind powder particles, later sintered for strength.	Sand, metal powders, ceramics, full-color sandstone	High speed, multi-color capability, cost-effective for volume	Lower strength until sintered, rough finish	Metal parts, casting molds, architectural models, prototypes
Material Jetting (PolyJet)	Droplets of photopolymer are jetted and cured with UV light; supports are added automatically.	Photopolymers, flexible resins, transparent resins	High resolution, multi-material, smooth surface	High cost, fragile parts, limited durability	High-resolution prototypes, medical models, concept models
EBM (Electron Beam Melting)	Electron beam melts metal powder layer by layer in a vacuum environment.	Titanium alloys, Inconel	High strength, fully dense parts, ideal for biomedical	High cost, slower than laser-based metal printing	Aerospace, orthopedic implants, high-strength metal parts
LOM (Laminated Object Manufacturing)	Layers of adhesive-coated paper, plastic, or metal laminates are cut and glued together using heat and pressure.	Paper, plastic sheets, composites	Low cost, large parts, environmentally friendly	Lower accuracy, rough surface finish, not for high-strength applications	Large concept models, patterns for casting, educational models

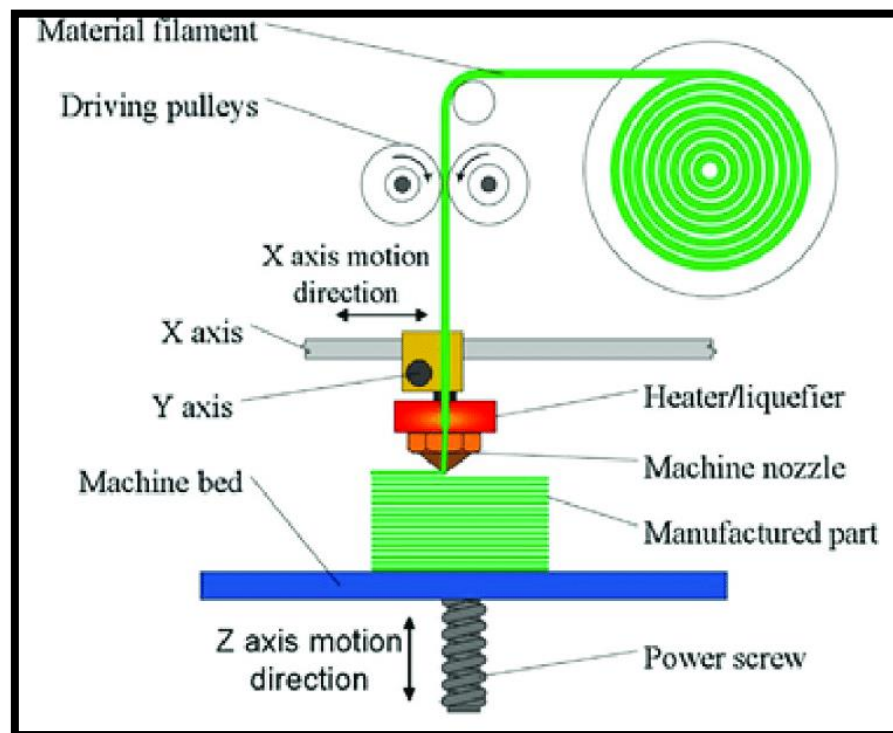
5.1 Fused Deposition Modeling (FDM)

FDM (Fused Deposition Modeling), also known as FFF (Fused Filament Fabrication), is the most common and affordable 3D printing technology.

The process works by melting plastic filament that is deposited, via a heated extruder, a layer at a time, onto a build platform according to the 3D data supplied to the printer. Each layer hardens as it is deposited and bonds to the previous layer.

The FDM processes require support structures for any applications with overhanging geometries. For FDM, this entails a second, water-soluble material, which allows support

structures to be relatively easily washed away, once the print is complete. Alternatively, breakaway support materials are also possible, which can be removed by manually snapping them off the part.



Working Principle

1. Design the 3D Model

- Create a 3D design using CAD software like Fusion 360, SolidWorks, or Tinkercad.
- Export the model as an STL or OBJ file.

1. Slicing the Model

- Use slicing software (e.g., Cura, PrusaSlicer) to convert the 3D model into G-code.
- The slicer sets parameters like layer height, infill density, speed, and support structures.

2. Heating and Extrusion

- The printer heats the nozzle (typically 180–260 °C, depending on material).
- Filament is fed through the extruder into the hot end, where it melts.

3. Layer-by-Layer Deposition

- The nozzle moves according to the G-code and deposits melted filament onto the build platform.
- Each layer cools and solidifies before the next one is deposited.

4. Cooling and Solidification

- A cooling fan ensures **strong layer adhesion** and **dimensional accuracy**.

5. Post-Processing

- Remove support structures if any.
- Sanding, painting, or chemical smoothing may be done for surface finishing.

Advantages of FDM

- Low cost and widely available
- Easy to operate and beginner-friendly
- Variety of materials and colours
- Suitable for rapid prototyping and functional parts

Limitations

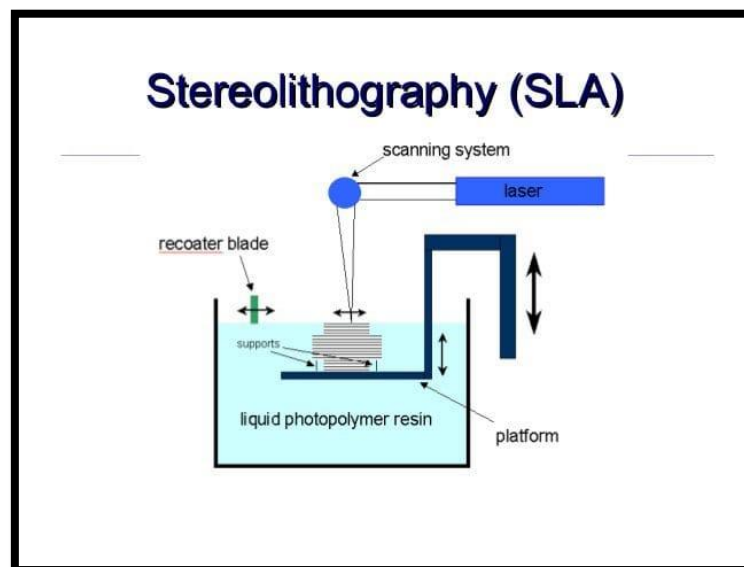
- Surface finish may require post-processing
- Limited mechanical strength compared to metal printing
- Lower accuracy compared to SLA or SLS
- Warping and layer separation if not properly calibrated

Applications

- Prototyping: Design testing and visualization
- Education: Teaching design and manufacturing concepts
- Functional Parts: Jigs, fixtures, and custom tools
- DIY Projects: Home gadgets, hobby models, and replacement parts

5.2 Stereolithography (SLA)

Stereolithography (SLA) is an additive manufacturing process that uses a UV laser to cure liquid photopolymer resin layer by layer, creating a 3D object from a 3D digital model. It is known for producing high-resolution parts with smooth surface finishes, sharp details, and high accuracy, making it ideal for prototypes, cosmetic models, and complex parts in fields like product design and medicine.



Working Principle-

- Stereolithography is a laser-based additive manufacturing process.
- It uses a vat of liquid photopolymer resin that solidifies when exposed to a UV laser.
- A laser beam is scanned across the resin surface in X–Y directions according to the digital 3D CAD file (usually in .stl format).
- The resin hardens precisely where the laser strikes.
- After one layer is cured, the platform lowers slightly along the Z-axis, and the next layer is traced.
- This process repeats layer by layer until the part is complete.
- Support structures are often required for overhangs or undercuts. These supports are removed after printing.
- Post-processing steps include cleaning (removing excess resin) and curing (hardening under UV light for final strength).

Materials Used

Photopolymer resins (UV-sensitive liquid polymers).

Types of SLA resins include:

- Standard resins (general purpose).
- Tough resins (stronger, impact resistant).
- Flexible resins (rubber-like).
- Transparent resins (optical applications).
- Dental/biocompatible resins (for medical uses).

Advantages

- High accuracy and excellent surface finish.
- Capable of producing complex geometries and detailed features.
- Useful for making high-quality prototypes with fine resolution.
- Smooth surfaces often reduce the need for extensive finishing.

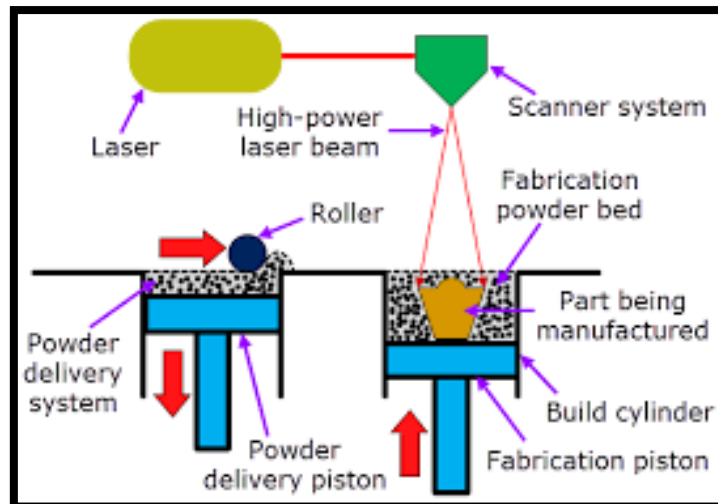
Limitations

- Requires support structures, which must be removed manually.
- Post-processing (cleaning, curing) is necessary.
- Photopolymer resins can be brittle and degrade over time.
- Machines and materials are relatively expensive compared to other 3D printing methods.
- Build size is limited compared to some other processes.

Applications

- Prototyping for product development.
- Automotive & aerospace industries for lightweight and precise models.
- Medical and dental fields – surgical guides, dental models, prosthetics.
- Consumer products – design validation, aesthetic models.
- Research & engineering – functional testing of designs.

5.3 Selective Laser Sintering (SLS)



Working Principle

- SLS is a powder-based additive manufacturing process.
- A thin layer of powder material (commonly nylon, polymers, or composites) is spread across the build platform.
- A high-power laser (CO₂ laser) scans across the powder bed and selectively fuses particles together according to the CAD model.
- After one layer is sintered, the platform lowers slightly along the Z-axis, and a new powder layer is spread on top.
- The process repeats layer by layer until the object is completed.
- Unlike SLA, no separate support structures are required because the surrounding unfused powder acts as support.
- Post-processing includes cooling, removing excess powder, and sometimes surface finishing (like polishing or coating).

Materials used

- Thermoplastic powders (most common: Nylon/PA12).
- Filled powders (glass-filled nylon, carbon fiber-reinforced nylon).
- Polystyrene powders (for investment casting patterns).
- Metals and ceramics (in advanced SLS variants – often called DMLS/SLM for metals).

Advantages

- No need for dedicated support structures.
- Can produce complex geometries and interlocking parts.
- Strong and functional prototypes can be created.
- Wide variety of engineering-grade materials available.
- High mechanical strength and durability of parts.
- Suitable for batch production of end-use parts.

Limitations

- Surface finish is usually rough and grainy compared to SLA.
- Post-processing (e.g., polishing, dyeing) may be needed for aesthetics.

- Cooling time is required before removing the part (to avoid warping).
- Machines are expensive and require controlled environments.
- Limited in producing highly transparent or very fine-detail parts compared to SLA.

Applications

- Functional prototypes in automotive, aerospace, and engineering.
- Low-volume production of end-use parts.
- Complex assemblies with interlocking parts.
- Medical devices such as orthotics and prosthetics.
- Consumer products – footwear, eyewear frames, customized goods.
- Rapid tooling for molds and fixtures.

6. Materials Used in 3D Printing

6.1 Plastics

- **Nylon (Polyamide):** Available in powder (sintering) or filament (FDM) form. It is strong, flexible, and durable, usually white but can be dyed before or after printing. In powder form, it can be blended with aluminium to create **Alumide**, used for lightweight automotive and aerospace parts.
- **ABS (Acrylonitrile Butadiene Styrene):** Popular for entry-level FDM printers. Strong, affordable, available in many colours, and widely used for prototypes, toys (e.g., LEGO bricks), and automotive components.
- **PLA (Polylactic Acid):** A biodegradable plastic, offered in resin (DLP/SL) and filament (FDM) form. Comes in many colours, including transparent, making it useful for display models and packaging. Less durable than ABS.
- **LayWood:** A wood–polymer composite filament, producing prints that look and smell like real wood, often used for decorative objects and furniture prototypes.

6.2 Metals

- **Stainless Steel:** Commonly used in powder form for sintering, melting, or EBM processes. Naturally silver but can be plated for decorative finishes. Used in tools, machine parts, and jewellery.
- **Aluminium & Cobalt Alloys:** Lightweight and strong, often applied in aerospace and automotive industries.
- **Titanium:** Extremely strong and corrosion-resistant, widely used for aerospace, medical implants, and high-performance engineering components.
- **Gold & Silver:** Directly printable in powder form, mainly for jewellery and luxury goods.

6.3 Ceramics

Ceramic materials can be 3D printed but require post-processing steps such as firing and glazing, similar to traditional ceramics. They are used for pottery, kitchenware, dental implants, and artistic applications.

6.4 Bio-Materials

Extensive research is being done on 3D-printed bio-materials. Living tissues are being explored for printing **organs** (for transplants), **prosthetics**, and **skin grafts**. Food printing is

also under development, with experimental printing of **meat substitutes** and other edible products.

7. Applications of 3D Printing

a) Industrial Applications

- **Prototyping** – Rapid iteration of designs
- **Tooling and Fixtures** – Custom jigs, molds, and fixtures
- **End-use parts** – Low-volume, high-customization products

b) Medical and Healthcare

- Custom **implants, prosthetics**, and **dental aligners**
- 3D-printed surgical models for preoperative planning

c) Aerospace and Automotive

- Lightweight components
- Complex parts impossible with traditional methods

d) Consumer Products

- Customized jewellery, footwear, home décor, and eyewear

e) Education and Research

- Models for visualization, experimentation, and hands-on training

8. Advantages of 3D Printing

3D printing confers several unique advantages over traditional manufacturing methods:

- Customised Production:** 3D printing easily creates customised products tailored to individual needs rather than mass-standardised products. It enables affordable personalisation and tool-less manufacturing.
- Complex Geometries:** Intricate shapes and complex internal structures like lattices, and honeycombs that enhance strength and functionality can be 3D printed. These are impossible to manufacture otherwise.
- On-Demand Local Production:** Digital inventories and distributed 3D printing reduce shipping costs, delays, carbon emissions and the need for global supply chains. Spare parts can be printed on-site.
- Material Savings:** Additive 3D printing deposits only required material minimising waste. Scrap can be recycled into filament, powder or feedstock.
- Democratised Manufacturing:** 3D printing grants access to affordable, quality manufacturing capabilities to individuals, startups, and small firms. Reduces barriers of expensive tools, moulds and dedicated factories.
- Rapid Design Iteration:** The ability to swiftly 3D print design prototypes enables faster innovation cycles, improvements and performance optimisation before finalising the design.

- g) **Education and Training:** 3D printing develops creative design thinking and spatial skills. Student engagement and understanding of concepts like design, engineering, arts, and mathematics improve.
- h) **Supply Chain Resilience:** On-site 3D printing of parts using digital inventories reduces supply chain disruptions due to geopolitical tensions, pandemics, and natural disasters. It strengthens localised manufacturing capacity.

9. Limitations of 3D Printing

Aspect	Limitation
Speed	Slower than mass production techniques
Cost	High for industrial-grade printers and materials
Material Properties	Limited mechanical strength for some polymers
Size Constraints	Limited build volume for most machines
Surface Finish	May need post-processing for smoothness
Technical Expertise	Requires knowledge of design, slicing, and machine operation

10. Future Trends

- Multi-material and multi-color 3D printing
- Use of **bioprinting** for organs and tissues
- **4D printing** – Materials that change shape over time
- Increased integration with **AI and automation** for process optimization